

Lab8

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(1) $\phi = (x_1 \vee x_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2 \vee x_2)$

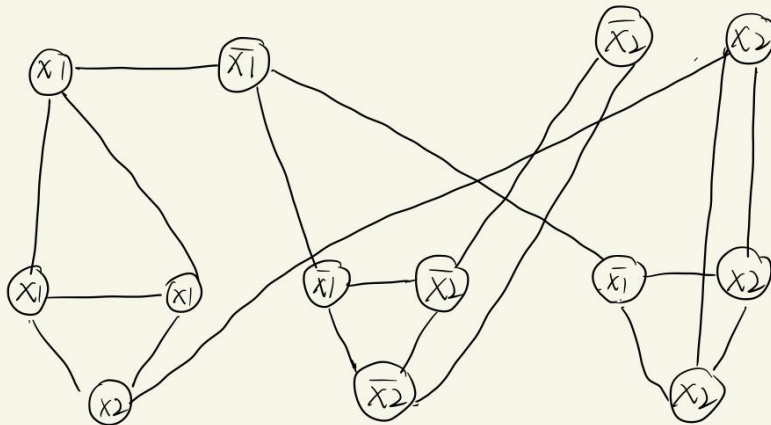
8 vector-cover

① Variable gadget

For each variable x in ϕ , we produce an edge connecting $x - \bar{x}$. If x is TRUE, select left node; whereas FALSE, select right node.

② Clause gadget

Each clause gadget is a triple of three nodes that are labeled with three literals of the clause. These three nodes are connected to each other and to the node in the variables gadgets that have identity labels.



Explanation:

① We put the nodes of the variable gadgets that correspond to the true literals into vertex cover.

② Then, we do ^{not select} one true literal in every clause and put the remaining two nodes from every clause gadget to the vertex cover.

$$\begin{array}{lll} C_1 = x_1 \vee x_1 \vee x_2, & C_2 = \bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_2, & C_3 = \bar{x}_1 \vee x_2 \vee x_2 \\ \quad = a_1 \vee a_2 \vee a_3 & \quad = b_1 \vee b_2 \vee b_3 & \quad = d_1 \vee d_2 \vee d_3 \\ \quad = x_1 \vee x_2 & \quad = \bar{x}_1 \vee \bar{x}_2 & \quad = \bar{x}_1 \vee x_2 \end{array}$$

C_1 is satisfiable iff x_1 is TRUE or x_2 is TRUE,

C_2 is satisfiable iff x_1 is FALSE or x_2 is FALSE,

C_3 is satisfiable iff x_1 is FALSE or x_2 is TRUE.

So C_1, C_2, C_3 is simultaneously satisfiable when x_1 is FALSE(F) and x_2 is TRUE(T).

x_1 is F, so we select $\bar{x}_1$ into the vertex cover.

x_2 is T, so select x_2 into the vertex cover.

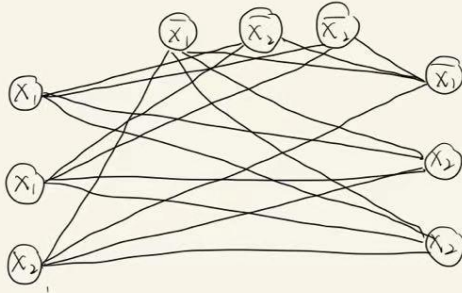
C_1 select a_1, a_2 . C_2 select b_1, b_3 . C_3 select d_2, d_3 .

vertex-cover = $\{ \bar{x}_1, x_2, a_1, a_2, b_1, b_3, d_2, d_3 \}$

$$(2) \quad \phi = (x_1 \vee \bar{x}_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee x_2 \vee x_3)$$

3-clique

- ① No edge is present between nodes in the same triple.
- ② No edge is present between nodes with contradictory labels, such as x_2 and $\bar{x}_2$.



$$\begin{array}{lll} C_1 = x_1 \vee x_1 \vee x_2 & C_2 = \bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3 & C_3 = \bar{x}_1 \vee x_2 \vee x_3 \\ \quad = a_1 \vee a_2 \vee a_3 & \quad = b_1 \vee b_2 \vee b_3 & \quad = d_1 \vee d_2 \vee d_3 \\ \quad = x_1 \vee x_2 & \quad = \bar{x}_1 \vee \bar{x}_2 & \quad = \bar{x}_1 \vee x_2 \end{array}$$

C_1, C_2, C_3 is satisfiable if $x_1 = F, x_2 = T$.

Rule ①: In each triple of G , we select one node corresponding to true literal. If more than one literal is true in clause, we choose one of the true literals arbitrarily.

C_1 select a_3 . C_2 select b_1 . C_3 select d_2 .

$$3\text{-CLIQUE} = \{a_3, b_1, d_2\}.$$